

Immersive Flora: Re-Engaging with the Forest Through the Visualisation of Plant-Environment Interactions in Virtual Reality

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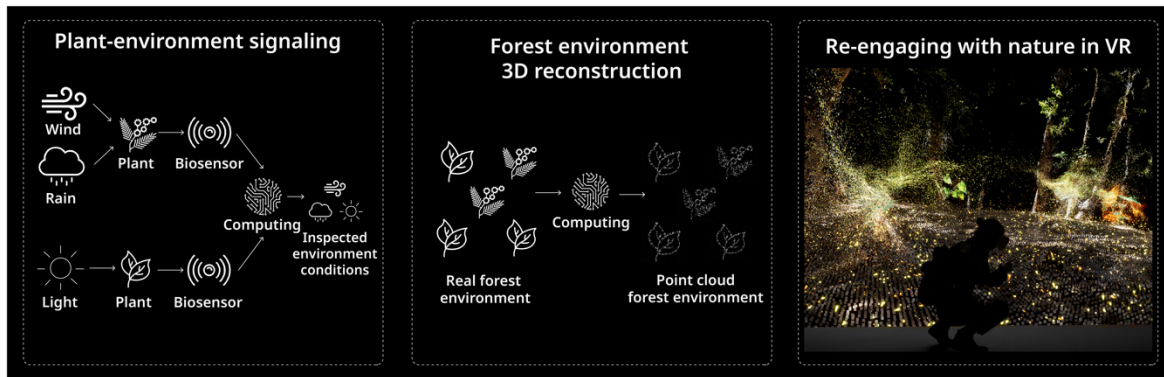


Figure 1: Description of Immersive Flora: A new medium to re-engage humans with the natural environment in the forest.

ABSTRACT

In response to recent calls for Human-Computer Interaction (HCI) to address ongoing environmental crises and promote the engagement between humans and nature, this paper presents Immersive Flora, a living plants-driven interactive system that re-engages users with the natural environment by virtually showcasing the diverse plant-environment interactions inside a forest. By utilizing biosensors to detect and analyze the intricate biological reactions of plants to wind, rain and sunlight, we explore the diversity of natural elements perceived by these plants within their wilderness habitat. This information is subsequently translated into a sophisticated virtual particle system that will react in real-time to mimic a change in the natural environment. This system is constructed utilizing point cloud data derived from the 3D reconstruction of plants and the forest environment. Through the expression of plant-environment interactions via particle manipulation, we establish an innovative interface reconnecting humans to the natural world.

CCS CONCEPTS

• Applied computing; • Media arts;; • Human-centered computing; • Virtual reality;

KEYWORDS

nature engagement, plant-environment signaling, virtual environment, immersive experience, more-than-human

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1 INTRODUCTION

Forests, hosting more than two-thirds of all terrestrial ecosystems, are currently facing an unprecedented and monumental challenge due to anthropogenic climate change and overuse through human intervention. This is resulting in alarming environmental issues, such as an increase in natural disturbances and a massive loss of biodiversity. In response, an increasing number of researchers are harnessing cutting-edge technologies to address these two challenges. Recent advancements in information and engineering technologies have facilitated the combination of remote sensing (RS) and artificial

intelligence (AI) for precise monitoring of environmental conditions [1] and biodiversity [2] within forests. While these scientific endeavors are undoubtedly essential, there remains an imperative need for the public to engage to counter these environmental issues actively [3]. In this context, new visualisation technologies can serve to facilitate the public's awareness and connection to the natural environment in forests.

Within this framework, HCI plays a crucial role in bridging the gap between humans and nature as it can facilitate technological interfaces that enable the public to enhance understanding, connection, and engagement with forest ecosystems [4]. Numerous projects have been proposed to re-engage the public with the forest environment using innovative approaches. For instance, establishing direct or indirect connections between users and the natural environment through digital sensory enhancement devices [5–7]. Other projects also represent the forest ecosystems by utilizing artificial sensors to gather environmental information for driving computer-generated visuals [8, 9]. Furthermore, creating immersive experiences for users within forest landscapes with the use of 3D scanning, modeling and extended reality XR technologies [10, 11]. These projects have established new interactive interfaces enabling novel connections between humans and forests. However, a significant portion of these endeavors relies on artificial sensors to collect and represent forest environmental information from a human-centered perspective, thereby overlooking the viewpoints of other species within the forest.

Beyond these novel human-centered interactive approaches, there has been an assertive call to explore the human-nature relationship from perspectives beyond the human-centered view. Stemming from ecophilosopher David Abram's "the more-than-human world" concept [12], an increasing amount of research has emphasized the potential significance of embracing a more-than-human perspective in technology development and design practice [13–16]. These provocations encourage us to engage in an entangled relationship with the natural world, recognizing its intricate web of life that extends beyond human boundaries. They prompt us to "understand" the perspectives, inherent value, and agency of other species, and invite us to cultivate a profound sense of wonder, respect, and reciprocity in our interactions with nature. Within HCI, researchers have responded to this call by creating novel interactions between humans and nonhumans to envision preferable futures that involve engaging with nature through an environmental awareness that extends beyond human boundaries, encompassing the perspectives of other species [17–20]. This expanded engagement promises new insights into our relationship with the natural world.

In response, plants have been progressively integrated into the HCI system in recent years because of their natural abilities to sense and respond to their environment [21, 22]. We further explore this field by focusing on the plant-environment interaction through the lesser-explored "embedded direct integration" technique [21]. By embedding plants with biosensors and employing machine learning algorithms, we monitor and analyze their bio-signal changes to predict environmental factors perceived by plants, including wind, rain, and sunlight. Furthermore, motivated by documented impacts on human behavior [23, 24], we incorporate immersive VR to represent these invisible biological processes of plants and the predicted environmental factors in human sensory ratios. This

decision aligns with the democratization of this technology, offering advantages such as making the invisible visible with computer graphic content and visual effects systems [25] and bringing distant natural environments to the user based on 3D scanning and modeling technologies [26, 27]. By linking the flora avatar and virtual environment with the interaction between living plants and the real environment, Immersive Flora enables users to immerse themselves in the perceptual environments of plants in the remote forest. By entangling with the more-than-human perspective, we expand our understanding of the natural world, promoting broader and deeper ecological connections, and consequently fostering better preservation and coexistence within our shared planet.

2 IMMERSIVE FLORA SYSTEM

Immersive Flora is a hybrid system that allows users to engage with the remote natural environment by virtually showcasing diverse plant-environment interactions within a forest. In this immersive experience, plants act as sensors, detecting changes in natural elements and seamlessly transmitting this information to a virtual environment that faithfully replicates reality and dynamically responds to alterations. As detailed in Figure 1 and subsequent subsections, this system consists of three stages: 1) plant-environment signaling, 2) forest environment 3D reconstruction, and 3) plant-environment interaction visualization in VR.

2.1 Plant-Environment Signaling

Throughout evolution, plants have developed the capacity to sense and respond to diverse environmental stimuli [28]. With advancements in electrophysiological technology, we can now precisely detect these responses by using bioamplifiers to monitor biological signal changes within plants, resulting from the transmission of chemical compounds within plant cells [29]. This technological integration with plant physiology deepens our understanding of how plants interact with their environment. Moreover, this approach not only broadens our scientific understanding but also offers potential applications in HCI [21]. In this project, we focus on detecting and analyzing the biosignal changes in plants in response to wind, rain, and sunlight stimuli. As shown in Figure 2a, we established an observation station in the natural environment. Within the observation station, we selected two types of plants as our sensing units: two *Cojoba arborea* plants, which are sensitive to mechanical stimuli such as wind and rain, and three plants from the *Araceae* family which are sensitive to light stimuli. To detect plant biosignals, a customized biosensor with two integrated amplifiers and a microprocessor is utilized. Figures 2b and 2c demonstrate the connection of the biosensor to the plants using two electrodes. For the wind and rain sensing unit, a reference electrode is inserted into the soil, and an Ag/AgCl electrode is placed on the plant's pulvinus with conductive gel applied. In the light sensing unit, two Ag/AgCl patch electrodes are attached to a single leaf of the plant. These biosensors detect subtle electrical potential signals from chemical reactions within the plants, and the biosignals are wirelessly transmitted to a Raspberry Pi running a program developed in openFrameworks for further analysis.

Figure 2d illustrates the configuration of the plant-environmental signaling system. In the analysis stage of the wind and rain sensing

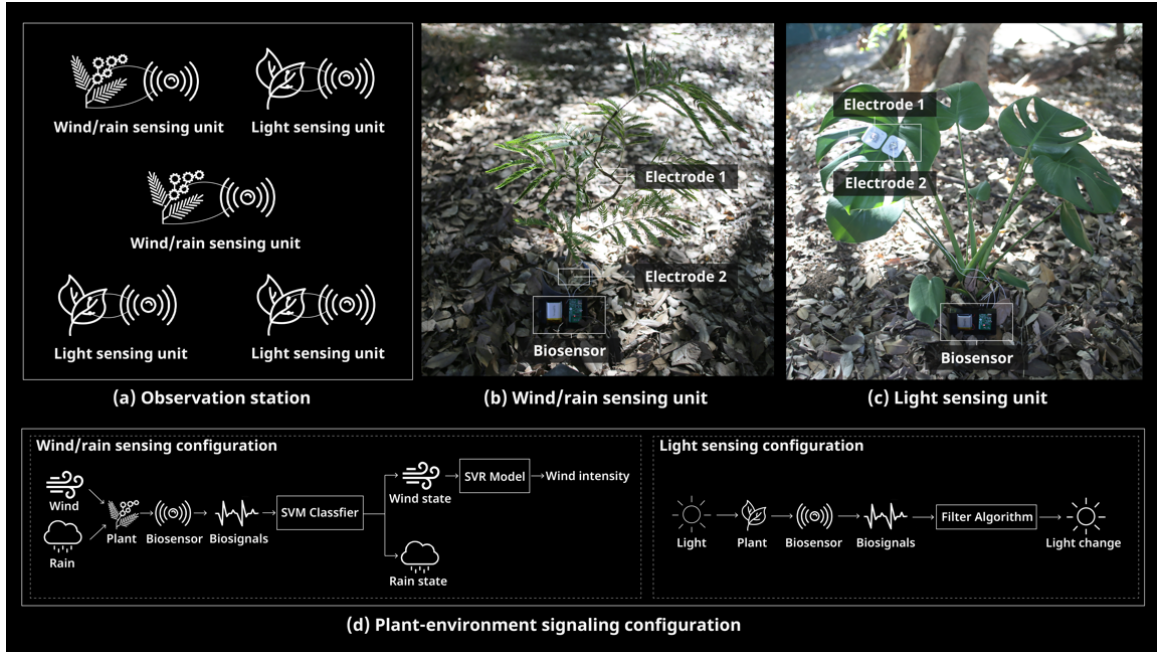


Figure 2: Illustration of plant-environment signaling: (a) observation station configuration, (b) wind and rain sensing unit, (c) light sensing unit, (d) plant-environment signaling configuration.

system, we applied two supervised learning algorithms to conduct a time-domain-based analysis of biosignals, predicting the biological feedback states of the plant under these two environmental stimuli. A non-linear Support Vector Machine (SVM) classifier was trained to distinguish between biosignal activities stimulated by wind and rain. Additionally, a non-linear Support Vector Regression (SVR) model was trained to predict wind intensity sensed by the plant as a continuous value between 0 and 255. In contrast to mechanical stimuli like wind and rain, plants exhibit slower biosignal changes in response to light stimuli. So during the analysis phase of the light sensing system, we first applied a Butterworth low-pass filter to isolate and retain the dynamic components of gradual changes associated with the plant's response to light. We then employed a sliding window analysis with a one-minute window size to calculate the intensity of signal variations within each minute (normalized as a value between 0 - 1), allowing for a representative measure of the environmental light conditions when plants experience changes in sunlight exposure. By combining these two types of sensing units, the system enables us to comprehend how plants respond to various environmental stimuli. The outcomes generated by the system will be represented to the users through a virtual environment in the subsequent steps, facilitating a comprehensive understanding of plant reactions to diverse environmental stimuli.

2.2 Forest Environment 3D Reconstruction

To fully immerse users in the virtual environment, we aimed to cover a forest plot of standard dimensions (approx. 20x20m) while capturing intricate details of the incorporated plants, including their delicate stems and leaves. While solid-state lidar sensors have

been introduced as a more economical alternative to the more sophisticated but prohibitively expensive terrestrial laser scanner TLS, they still lack sufficient coverage and suffer from poor geometric accuracy in the acquired point cloud [30]. Therefore, we chose to use a combination of image-based survey techniques, providing complete coverage of the entire forest scene and a high-resolution dense point cloud for the plant avatars.

For the 3D reconstruction of the forest, spherical photogrammetry coupled with 360 cameras has recently proven to be the optimal choice for capturing forest environments [31]. We followed a similar approach, walking at a constant pace with a 360 camera (ONE X2, Insta360, Shenzhen, China) mounted on the operator's head along the path outlined in Figure 3b. This resulted in two 360-degree sphere videos, loaded into the photogrammetric software Agisoft Metashape and converted into 70 frames and 80 frames, respectively. Subsequently, we followed the standard photogrammetric workflow, obtaining two dense point clouds merged with coded targets' assistance (Figure 3e).

To achieve a finer level of precision in the 3D reconstruction of plants connected to sensors, we used terrestrial close-range photogrammetry with a digital single-lens reflex (DSLR) camera (5D Mark III, Canon, Tokyo, Japan). As shown in Figure 3a, a convergent network was established for each plant, with between 50 to 80 photos taken per plant. In front of each plant, a coded-target pair was placed, with the distance from both centers known (12.9 cm). This facilitated precise scaling (Figure 3c) and alignment for the accurate representation of both individual plants and their integration into the virtual environment captured with the 360 camera (Figure 3f). Finally, the point clouds were saved in .ply

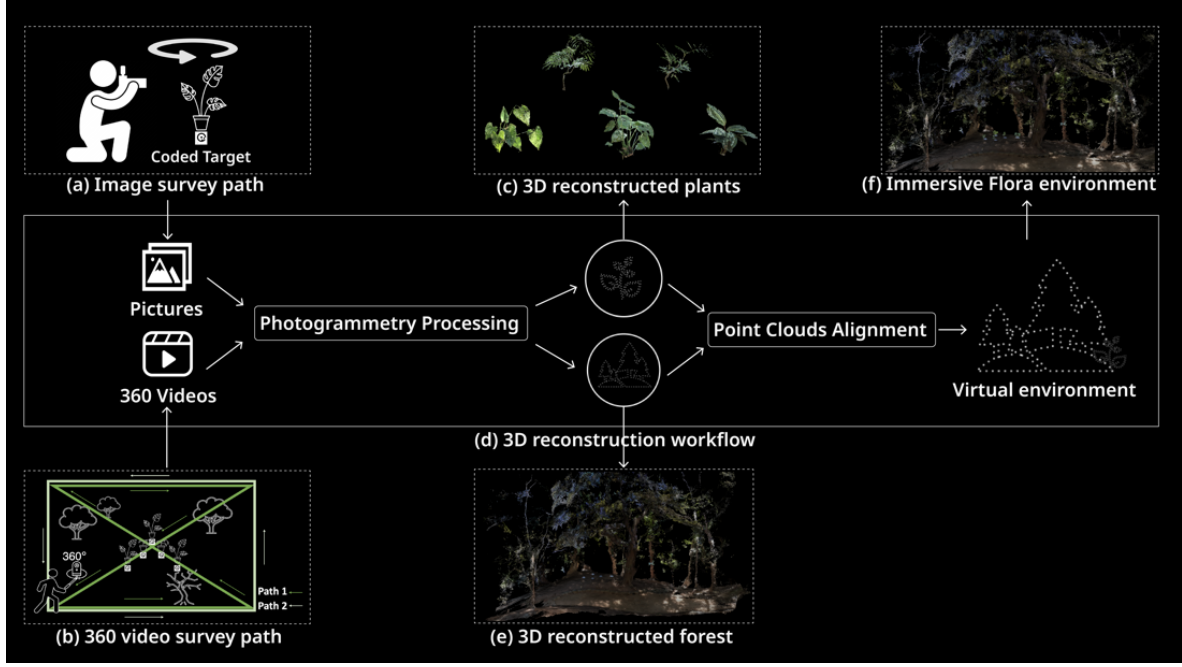


Figure 3: Illustration of virtual environment 3D reconstruction procedure. (a) image and (b) 360 video survey paths, 3D reconstructed (c) plants and (e) forest, (d) 3d reconstruction workflow, (f) Immersive Flora environment

format, enabling the import of 3D point clouds into Unity using the PCX plugin [32].

2.3 Plant-Environment Interaction Visualization in VR

Following the Plant-Environment Signaling and Virtual Environment 3D Reconstruction stages, we designed a Unity-based VR system to visualize the natural environment stimuli perceived by multiple plants. To represent these invisible biological processes of plants, we customized three distinct particle effects based on the visual effect package of Unity. As depicted in Figure 4, each plant's 3D point cloud was associated with particle effects using point cloud position and color to initialize every particle. These particle effects mirror rain, wind, and light factors, with each being controlled by the monitored environmental conditions. The analysis results from the plant-environment signaling systems (2.1) are wirelessly transmitted through messages queuing telemetry transport (MQTT) to the Unity system. This prompts an immediate update in the system's visual appearance, aligning with one of the following particle effects.

Rain particle effect: creates a dynamic motion resembling the flow of rainwater, applying downward forces and subtle turbulent forces. When the plant rain sensing units detect rain conditions from the SVM model, it sends an event message to trigger an increase in the number of rain particles for a visually enhanced representation of the rained environment in the remote forest.

Wind particle effect: emulates the flow of the wind field in a natural pattern. We employ a Perlin noise-type turbulence force to stimulate the movement of particle clusters. The wind intensity,

represented as a continuous integer value ranging from 0 to 255, calculated by the plant's wind sensing unit through the SVR model, is mapped to both the force intensity (ranging from 0 to 0.2) and the frequency (ranging from 2 to 10) of the Perlin noise.

Light particle effect: simulates a natural light field on the ground caused by the movement of sunlight during the day. We use a turbulence force to drive the light particle's movement. Through three plant light sensing units, we control the flow state and intensity of three light particle effects around their respective positions. Changes in light exposure sensed by the units inspected by the intensity of biosignal variations (ranging from 0 - 1) within each minute mapped to both the force intensity (ranging from 0 to 0.3) and the red channel of color attributes (ranging from 0 to 255).

This VR system surpasses mere digital simulation by creating an immersive experience with the help of particle effects triggered by plants' biosignals. It serves as a bio-hybrid virtual medium that connects nature and humans. From a plant's perspective, it becomes a unique means of communicating their perception of climate change impact within the forest. In interactive sessions, users can enter the Immersive Flora world through a VR headset, navigating a virtual forest populated with 5 plant avatars. Changes in particle fields, driven by plant-environment interactions, allow users to reconnect with various natural elements perceived by the plants.

3 CONCLUSION AND FUTURE WORKS

In conclusion, our work introduces a novel method for humans to gain awareness of the environmental impact on forests by providing insights into plant perspectives. This involves translating the

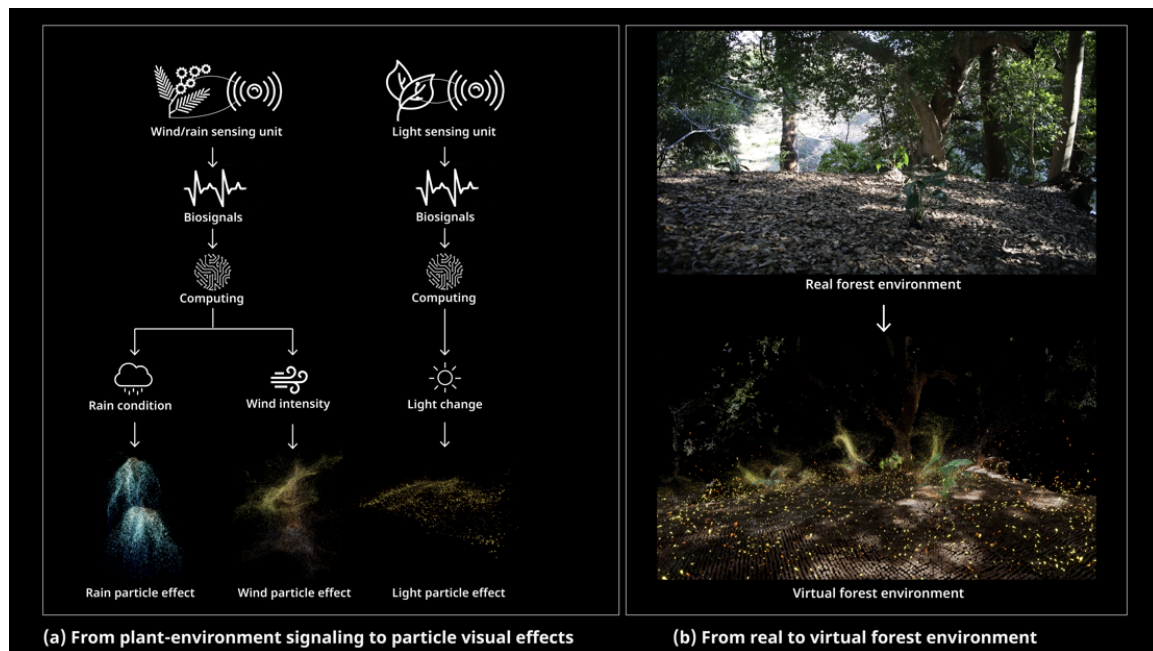


Figure 4: Description of plant-environment interaction visualization: (a) from plant-environment signaling to particle visual effects, (b) from real to virtual forest environment.

often-unseen biological processes of plants into human sensory experiences using engineering and intelligent computing. The result is a reconnection between humans and nature within the perceptual worlds of other species. Our work deepens users' understanding of the delicate equilibrium in our forests, which are currently at a breaking point. Through the varied perspectives of plants, our innovative approach enables users to immerse themselves in a remote forest environment, fostering a greater appreciation for the natural world. Immersive Flora, concentrating on rain, wind, and sunlight, presents a versatile platform for potential expansion, additional units could reproduce a more vivid and detailed representation of the natural environment from the plant's perspective. Furthermore, quantitative data on humans' ecological perception is lacking, prompting plans for a user study. If presented at SIGCHI, this offers an opportunity for data collection and feedback, advancing our immersive approach for users to explore remote forests through plant perspectives and bolster ecological preservation efforts.

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